

Module 1: reduced resistive MHD model

The project uses a deliberately small parent model before making any M3D-C1 claim:

$$(x, y, t) \in [0, 2\pi) \times [0, 2\pi) \times [0, T], \quad \mathbf{q} = (\psi, U), \quad \boldsymbol{\mu} = (\eta, \nu, A_0).$$

- ▶ Two-dimensional Cartesian domain: all resolved variation is in (x, y) ; no z dependence is solved.
- ▶ Constant density plus incompressibility: the flow is represented by one stream function U .
- ▶ Strong guide-field ordering: the total magnetic field is a large uniform $B_g \hat{\mathbf{z}}$ plus a smaller in-plane perturbation $\mathbf{B}_\perp$.
- ▶ Version-1 trajectories are synthetic reduced-MHD data, not direct M3D-C1 output.

$$\text{goal: } (x, y, t, \eta, \nu, A_0) \mapsto (\psi, U), \quad \text{then derive } J, \omega, \mathbf{B}_\perp, \mathbf{v}_\perp.$$

Two scalar potentials encode the divergence-free physical fields:

Magnetic perturbation

$$\mathbf{B}_{\perp} = \nabla\psi \times \hat{\mathbf{z}} = (\psi_y, -\psi_x), \quad \nabla \cdot \mathbf{B}_{\perp} = 0.$$

$$J = -\nabla^2\psi, \quad \text{out-of-plane current density.}$$

Center-of-mass flow

$$\mathbf{v}_{\perp} = \hat{\mathbf{z}} \times \nabla U = (-U_y, U_x), \quad \nabla \cdot \mathbf{v}_{\perp} = 0.$$

$$\omega = \nabla^2 U, \quad \text{out-of-plane vorticity.}$$

$$\mathbf{B} = B_g \hat{\mathbf{z}} + \mathbf{B}_{\perp}, \quad B_g = \text{constant background}, \quad \psi = \text{evolving perturbation flux.}$$

Only (ψ, U) are independent neural outputs. Penalizing or reporting J and ω tests derivative fidelity, not extra labels.

$$[f, g] = f_x g_y - f_y g_x$$

The stream-function velocity turns material advection into a Poisson bracket:

$$\mathbf{v}_\perp \cdot \nabla f = (-U_y) f_x + (U_x) f_y = [U, f], \quad \frac{Df}{Dt} = f_t + [U, f].$$

Algebraic properties

$$[f, g] = -[g, f], \quad [f, f] = 0.$$

$$\int_\Omega h [f, g] \, dA = \int_\Omega f [g, h] \, dA \quad (\text{periodic}).$$

Physical reading

- ▶ U labels the pattern of fluid circulation.
- ▶ ψ labels in-plane magnetic contours.
- ▶ Brackets move contours; diffusion changes them.

Nonlinear brackets redistribute structure and exchange magnetic/kinetic energy; the explicit dissipators set irreversible loss.

$$\psi_t + [U, \psi] = \eta \nabla^2 \psi = -\eta J$$

$$\omega_t + [U, \omega] = [J, \psi] + \nu \nabla^2 \omega, \quad \omega = \nabla^2 U, \quad J = -\nabla^2 \psi.$$

- ▶ $[U, \psi]$: magnetic-flux contours are carried by the center-of-mass flow.
- ▶ $\eta \nabla^2 \psi$: resistive diffusion lets plasma and field lines slip; this is where topology can change.
- ▶ $[U, \omega]$: vorticity transport by the same incompressible velocity field.
- ▶ $[J, \psi]$: reduced form of magnetic forcing from $\mathbf{J} \times \mathbf{B}$.
- ▶ $\nu \nabla^2 \omega$: viscous smoothing of shear and vorticity.

The two equations are closed because $(J, \omega, \mathbf{B}_\perp, \mathbf{v}_\perp)$ are derivatives of (ψ, U) .

Use Alfvén scales before any machine-learning standardization:

$$V_A = \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \quad t_A = \frac{L_0}{V_A}, \quad x = \frac{x_d}{L_0}, \quad t = \frac{t_d}{t_A}.$$
$$\eta = \frac{\eta_m}{L_0 V_A}, \quad \nu = \frac{\nu_d}{L_0 V_A}, \quad S = \eta^{-1}, \quad \text{Re} = \nu^{-1}, \quad \text{Pm} = \frac{\nu}{\eta}.$$

- ▶ $S = t_\eta/t_A$ compares resistive diffusion to Alfvénic motion; large S means slowly diffusing magnetic flux.
- ▶ $\text{Re} = t_\nu/t_A$ compares viscous diffusion to flow advection; large Re permits sharp velocity gradients.
- ▶ $\text{Pm} = \nu/\eta$ controls whether current sheets and vorticity sheets smooth on comparable scales.

physical nondimensionalization $\neq$ neural standardization $(q - m_q)/s_q$.

Module 1: boundary, gauge, and energy checks

Periodic faces and all PDE-required derivatives match:

$$q(0, y, t) = q(2\pi, y, t), \quad q(x, 0, t) = q(x, 2\pi, t), \quad q \in \{\psi, U, J, \omega\}.$$

The stream-function gauge and magnetic mean condition are

$$\boxed{\langle U \rangle(t) = 0}, \quad \boxed{\frac{d}{dt} \langle \psi \rangle = 0}.$$

The continuous periodic energy law is

$$\boxed{\frac{d}{dt} \frac{1}{2} \int_{\Omega} (|\nabla \psi|^2 + |\nabla U|^2) \, dA = -\eta \int_{\Omega} J^2 \, dA - \nu \int_{\Omega} \omega^2 \, dA \leq 0.}$$

A valid solver or PINN should not create energy from the bracket terms. Energy defects isolate sign errors, derivative errors, and broken periodicity.

Module 2: verified trajectory generation

For each parameter vector, Module 2 produces an accepted numerical realization

$$\boldsymbol{\mu} = (\eta, \nu, A_0), \quad \mathcal{Q}_{\boldsymbol{\mu}}^{N_x, N_y, \Delta t} = \{\psi, U, J, \omega\}_{nij}.$$

The advanced Fourier state is only

$$\hat{\mathbf{q}}(t) = \begin{bmatrix} \hat{\psi}(t) \\ \hat{\omega}(t) \end{bmatrix}, \quad \hat{U} = -\frac{\hat{\omega}}{k^2} \ (k \neq 0), \quad \hat{J} = k^2 \hat{\psi}.$$

- ▶ Zero modes enforce $\hat{U}_{00} = \hat{\omega}_{00} = 0$ and preserve the gauge/compatibility conditions.
- ▶ (U, J) are reconstructed inside every RK stage, not merely after the step.
- ▶ A run becomes training data only after operator tests, trajectory checks, convergence evidence, and acceptance flags.

The baseline periodic initial condition is

$$\boxed{\psi_0 = \sin x \sin y}, \quad \boxed{U_0 = A_0(\cos x - \cos y)}, \quad J_0 = 2\psi_0, \quad \omega_0 = -U_0.$$
$$\mathbf{v}_{\perp,0} = (-A_0 \sin y, -A_0 \sin x).$$

The flow drives like-signed current islands toward the central X-point.

approach $\rightarrow$ compression $\rightarrow$ current sheet

$\rightarrow$ resistive reconnection $\rightarrow$ merged topology.

- Baseline: $\eta = \nu = 10^{-3}$, $A_0 = 0.1$, $N_x = N_y = 128$, $\Delta t = 10^{-3}$, $T \approx 8$.
- The case exercises all terms: bracket transport, magnetic forcing, resistive diffusion, and viscous diffusion.
- The horizon must cover the first resolved peak and a post-peak interval, not just early smooth motion.

Use the half-open grid

$$x_j = 2\pi j/N_x, \quad y_\ell = 2\pi \ell/N_y, \quad j = 0, \dots, N_x - 1, \quad \ell = 0, \dots, N_y - 1.$$

No endpoint is duplicated, so periodic differentiation is algebraic:

$$\widehat{f}_x = ik_x \widehat{f}, \quad \widehat{f}_y = ik_y \widehat{f}, \quad \widehat{\nabla^2 f} = -k^2 \widehat{f}.$$

Linear pieces

$$\widehat{U} = -\widehat{\omega}/k^2, \quad \widehat{J} = k^2 \widehat{\psi}.$$

They are exact for resolved modes.

This Fourier–grid–Fourier loop defines the pseudo-spectral method.

Nonlinear pieces

$$[f, g] = f_x g_y - f_y g_x.$$

Derivatives are spectral; products are evaluated on the grid.

Quadratic products can alias unresolved high modes into false low modes. The frozen nonlinear operator is

$$\widehat{\mathcal{B}(f, g)} = \mathcal{P}_{2/3} \mathcal{F}(f_x g_y - f_y g_x), \quad |m| \leq \lfloor N_x/3 \rfloor, \quad |n| \leq \lfloor N_y/3 \rfloor.$$

The semi-discrete dynamics are advanced by classical RK4:

$$\mathbf{q}^{n+1} = \mathbf{q}^n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \quad \mathbf{k}_i = \mathcal{L}(\text{stage state}).$$

At each stage:

$(\psi, \omega) \rightarrow (U, J) \rightarrow \text{spectral derivatives} \rightarrow \text{grid products} \rightarrow \mathcal{P}_{2/3} \rightarrow \text{zero-mode enforcement}.$

Stability monitors are useful, but they do not replace explicit temporal refinement.

Energy balance is monitored through the integrated defect

$$\delta_E(t) = \frac{\left| E(t) - E(0) + \int_0^t [\eta \int J^2 dA + \nu \int \omega^2 dA] ds \right|}{\max(E(0), \epsilon_{\text{mach}})}.$$

Topology is summarized by O/X-point flux difference:

$$\Delta\psi_{OX} = \psi_X - \psi_O, \quad \Phi_{\text{rec}}(t) = \Delta\psi_{OX}(0) - \Delta\psi_{OX}(t), \quad \mathcal{R}_{\text{rec}} = \frac{d\Phi_{\text{rec}}}{dt}.$$

Additional acceptance quantities include

$$\|J\|_\infty, \quad \text{sheet width in grid cells}, \quad \sum_{\text{tail}} |\hat{q}_k|^2, \quad \langle U \rangle, \langle \psi \rangle, \quad \|J + \nabla^2 \psi\|, \quad \|\omega - \nabla^2 U\|.$$

A visually plausible contour plot is not enough.

Module 2: acceptance before data release

Verification proceeds from operators to full trajectories:

operators → analytic fields → bracket identities → exact decay
→ manufactured solution → energy balance → refinement
→ Python/MATLAB parity → case acceptance.

- ▶ Temporal refinement should show fourth-order behavior until spatial error dominates.
- ▶ Spatial refinement should stabilize peak time, reconnection history, sheet width, spectra, and derived fields.
- ▶ Accepted files store fields, coordinates, parameters, diagnostics, config, revision, host metadata, and flags.
- ▶ Failed or marginal runs can be retained as diagnostics, but not silently mixed into the canonical training release.

Data quality rule: unverified simulation output is not a reference solution.

Module 3: physical case versus numerical realization

Module 3 preserves identity so later model comparisons are auditable.

Physical case

`physical_case_id`

Equations, domain, initial/boundary conditions, physical parameters, and intended horizon.

Numerical realization

`case_id`

Solver, grid, time step, precision, language, configuration, code revision, and verification outcome.

$\text{release} \supset \text{physical cases} \supset \text{realizations} \supset \text{snapshots} \supset \text{grid points}.$

- ▶ Refined, restart, parity, or migrated realizations remain linked to the same physical case.
- ▶ Splitting by snapshots or grid points would leak a trajectory into both training and testing.
- ▶ The contract protects the distinction between “M3D-C1-shaped” and “M3D-C1-derived.”

Module 3: canonical HDF5 schema

Every accepted case declares

`schema_name=pinn_m3dc1.case,` `schema_version=1.0.0,` `axis_order=t,x,y.`

Arrays

`coordinates/` $\{t, x, y\}$
`fields/` $\{\psi, U, J, \omega\}$
`parameters/` $\{\eta, \nu, A_0\}$
`diagnostics/` $E, \delta_E, \Phi_{\text{rec}}, \dots$

Metadata

`physics/` equations, units, gauges
`numerics/` grid, Δt , solver
`verification/` tests and flags
`provenance/` code, config, host

$q[n, i, j] = q(t_n, x_i, y_j),$ $q \in \{\psi, U, J, \omega\},$ reference arrays are finite float64 values.

A non-square, non-symmetric sentinel detects silent Python–MATLAB axis transposes.

Module 3: whole-case splits and leakage control

Generalization is measured on disjoint physical-case sets:

$$\mathcal{P}_{\text{all}} = \mathcal{P}_{\text{tr}} \cup \mathcal{P}_{\text{va}} \cup \mathcal{P}_{\text{ti}} \cup \mathcal{P}_{\text{te}}, \quad \mathcal{P}_a \cap \mathcal{P}_b = \emptyset \quad (a \neq b).$$

- ▶ Training cases update weights and fit preprocessing; validation selects checkpoints and hyperparameters.
- ▶ Interpolation tests lie inside the training convex hull; extrapolation tests lie outside it.
- ▶ All realizations sharing one `physical_case_id` stay in one partition.
- ▶ The split manifest is frozen before model development and contains random seed, date, rule, and case lists.

$\hat{\alpha} = \text{FitTransform}(\mathcal{D}_{\text{tr}})$, then apply $\hat{\alpha}$ to validation/test data without refitting.

Refitting normalization or PCA-like transforms on held-out trajectories leaks field information.

Module 3: learning views leave truth immutable

The canonical release stores physical arrays; learning views are reproducible derived products.
For

$$(x, y, t, \eta, \nu, A_0) \mapsto (\psi, U),$$

periodic coordinates are encoded as

$$\mathbf{e}_{\text{per}} = (\sin \phi_x, \cos \phi_x, \sin \phi_y, \cos \phi_y), \quad \phi_x = 2\pi x/L_x, \quad \phi_y = 2\pi y/L_y.$$

Positive transport coefficients can be transformed by

$$r_\eta = \log_{10} \eta, \quad r_\nu = \log_{10} \nu, \quad \tilde{q} = \frac{q - m_q}{s_q}, \quad q = m_q + s_q \tilde{q}.$$

PINN derivatives require recorded chain-rule factors:

$$\partial_x q = \frac{s_q}{s_x} \partial_x \tilde{q}, \quad \partial_x^2 q = \frac{s_q}{s_x^2} \partial_x^2 \tilde{q}, \quad \partial_t q = \frac{s_q}{s_t} \partial_t \tilde{q}.$$

Module 3: protecting the M3D-C1 claim

Validation is layered:

structure $\rightarrow$ relations $\rightarrow$ numerics $\rightarrow$ physics $\rightarrow$ provenance $\rightarrow$ release.

Readers check axes, finite values, gauges, derived-field identities, diagnostics, checksums, disjoint splits, parent links, and migration history.

M3D-C1-shaped = metadata + mesh/coordinates + parameters
+ fields + diagnostics + splits + provenance.

- ▶ Unknown major schema versions are rejected; migrations create traceable new files.
- ▶ Genuine M3D-C1 exports remain source-native; adapters must document geometry, units, signs, gauges, projections, closures, and information loss.
- ▶ The project can claim interface readiness before it can claim physical equivalence to M3D-C1.

The baseline neural surrogate approximates the parameterized field family

$$\mathcal{N}_{\theta} : (x, y, t, \eta, \nu, A_0) \mapsto (\hat{\psi}, \hat{U}).$$

Complete cases, not individual points, define the split:

$$\mu_c = (\eta_c, \nu_c, A_{0,c}), \quad \mathcal{D}_c = \{(x_i, y_j, t_n, \mu_c; \psi, U)\}_{nij}.$$

- ▶ The model sees labels only through sampled field values of (ψ, U) .
- ▶ No PDE residual, derivative target, energy law, current/vorticity label, or boundary penalty appears in training.
- ▶ Time interpolation within $[0, T]$ is not forecasting beyond the simulated horizon.
- ▶ This baseline is the controlled comparator for asking whether physics information helps.

Periodic coordinates respect the domain before the network sees them:

$$\phi_x = \frac{2\pi x}{L_x}, \quad \phi_y = \frac{2\pi y}{L_y},$$
$$\mathbf{z} = (\sin \phi_x, \cos \phi_x, \sin \phi_y, \cos \phi_y, \tilde{t}, \tilde{r}_\eta, \tilde{r}_\nu, \tilde{A}_0).$$

- Smooth functions of $(\sin \phi, \cos \phi)$ are periodic in value and derivatives.
- $r_\eta = \log_{10} \eta$ and $r_\nu = \log_{10} \nu$ make decade-scale transport coefficients easier to learn.
- Scaling statistics are fit only on training cases.
- Baseline architecture: six fully connected tanh layers of width 128 plus a linear two-output head.

Module 5 retains the same architecture family and representation so the comparison isolates the training objective.

With standardized errors

$$e_{\psi,n} = \widehat{\tilde{\psi}}_n - \tilde{\psi}_n, \quad e_{U,n} = \widehat{\tilde{U}}_n - \tilde{U}_n,$$

the case-balanced objective is

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{data}} = \frac{1}{|\mathcal{C}_{\mathcal{B}}|} \sum_{c \in \mathcal{C}_{\mathcal{B}}} \left[\frac{1}{N_c} \sum_{n \in c} e_{\psi,n}^2 + \frac{1}{N_c} \sum_{n \in c} e_{U,n}^2 \right].$$

- Case balancing prevents long trajectories or dense grids from dominating the objective.
- Adam, mini-batches, early stopping, and regularization are numerical choices, not physics information.
- Any good behavior in J , ω , residuals, or energy must emerge indirectly from fitting (ψ, U) .

Module 4: evaluation makes the PINN testable

On complete unseen cases, report field and derivative-field errors:

$$\varepsilon_{L_2}^{(a)} = \frac{\|\hat{a} - a\|_2}{\|a\|_2 + \epsilon}, \quad a \in \{\psi, U, J, \omega\}, \quad \hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U}.$$

Diagnostics are evaluation-only for Module 4:

$$R_\psi, \quad R_\omega, \quad E(t), \quad \delta_E(t), \quad \Phi_{\text{rec}}(t), \quad \text{periodicity defects}, \quad \langle \hat{U} \rangle.$$

Controlled comparison rule: Modules 4 and 5 share splits, labels, preprocessing, periodic encoding, architecture family, data loss, metrics, seeds, and reporting windows.

Physics helps only if a declared held-out metric improves, with extra compute disclosed.

The network still predicts only $(\hat{\psi}, \hat{U})$, but automatic differentiation constructs

$$\hat{J} = -\nabla^2 \hat{\psi}, \quad \hat{\omega} = \nabla^2 \hat{U}.$$

The local reduced-MHD defects are

$$R_{\psi} = \hat{\psi}_t + [\hat{U}, \hat{\psi}] - \eta \nabla^2 \hat{\psi},$$

$$R_{\omega} = \hat{\omega}_t + [\hat{U}, \hat{\omega}] - [\hat{J}, \hat{\psi}] - \nu \nabla^2 \hat{\omega}.$$

- Labeled points constrain agreement with accepted trajectories.
- Collocation points constrain the equation even where no solver label is used.
- The residuals must be computed in physical coordinates after undoing scaling through the chain rule.

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{PDE}}\mathcal{L}_{\text{PDE}} + w_{\text{IC}}\mathcal{L}_{\text{IC}} + w_{\text{BC}}\mathcal{L}_{\text{BC}} + w_{\text{global}}\mathcal{L}_{\text{global}}.$$

Local terms

$$\mathcal{L}_{\text{PDE}} = \left\langle R_{\psi}^2/\sigma_{\psi}^2 + R_{\omega}^2/\sigma_{\omega}^2 \right\rangle_{\text{colloc}}$$

$\mathcal{L}_{\text{data}}$ = same labeled loss as Module 4.

Condition terms

- ▶ $\mathcal{L}_{\text{IC}}$ selects the intended trajectory at $t = 0$.
- ▶ $\mathcal{L}_{\text{BC}} = 0$ for hard periodic inputs; otherwise it measures face mismatch.
- ▶ $\mathcal{L}_{\text{global}}$ contains gauge, mean-flux, and optional balance constraints.

Weights are hyperparameters; they must be reported with normalization choices and collocation counts.

Because $\hat{\omega} = \nabla^2 \hat{U}$,

$$\hat{\omega}_t \sim \partial_t \partial_x^2 \hat{U}, \quad [\hat{U}, \hat{\omega}] \sim \partial_x^3 \hat{U}, \quad \nabla^2 \hat{\omega} = \nabla^4 \hat{U}.$$

The Lorentz bracket includes third derivatives of $\hat{\psi}$ through $\hat{J} = -\nabla^2 \hat{\psi}$.

- Smooth activations such as tanh are required; ReLU-like nonsmoothness is incompatible with fourth derivatives.
- Scaling factors must remain in the differentiable graph; otherwise residual units and magnitudes are wrong.
- Automatic-differentiation derivatives should be tested against analytic periodic functions before training.

The PDE does not identify a spatially constant offset in U :

$$\langle \hat{U} \rangle(t, \mu) = 0.$$

A useful optional balance penalty is based on the exact energy law:

$$\frac{d}{dt} \frac{1}{2} \int_{\Omega} \left(|\nabla \hat{\psi}|^2 + |\nabla \hat{U}|^2 \right) dA + \eta \int_{\Omega} \hat{J}^2 dA + \nu \int_{\Omega} \hat{\omega}^2 dA = 0.$$

Collocation defines what “PDE accuracy” means:

$$\mathcal{X}_{\text{colloc}} = \{(x, y, t, \eta, \nu, A_0)\}.$$

Fixed validation collocation and adaptive training collocation are different objects. Practical curriculum:

derivative tests $\rightarrow$ data warm start $\rightarrow R_{\psi} \rightarrow R_{\omega}$

$\rightarrow$ longer windows $\rightarrow$ multi-parameter cases.

Global penalties should follow quadrature verification; otherwise they can enforce a numerically wrong balance.

Compare matched models on unseen cases using

$$\varepsilon_{L_2}^{(\psi, U, J, \omega)}, \quad R_\psi, R_\omega, \quad \delta_E, \\ \Phi_{\text{rec}}, \mathcal{R}_{\text{rec}}, \quad \text{training and inference cost.}$$

A PINN can help by improving labeled-data efficiency, derivative fidelity, parameter generalization, or physical admissibility.

It can also fail through slow training, gradient conflict, oversmoothed sheets, residual imbalance, or wrong/incomplete physics.

Conclusion standard: the PINN is not assumed to win.

The result remains a surrogate of synthetic two-field reduced-MHD dynamics.

Full-code surrogate claim: genuine M3D-C1 data plus a validated adapter.